

Clinical Context Matters: Metadata-Conditioned Vision–Language Modeling for Pediatric Chest CT

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Abstract

Vision–language models for chest CT are trained and benchmarked almost exclusively on adult cohorts, and they read a volume with generic pathology prompts even though the clinical indication, age, and sex that motivated the examination are available at interpretation time. Both choices are costly in pediatric radiology, where anatomy, disease priors, and reconstruction geometry differ from adults and where indications are unusually informative. We present METAQ-CT, a metadata-conditioned Q-Former on an anatomy-routed 3-D CT encoder that retains a metadata-independent query bank while adding pathology-specific conditioned queries attending to indication, age, and sex through gated cross-attention and identity-initialized feature-wise modulation. On a frozen, MRN-disjoint pediatric test split of 1,273 studies spanning ages 0–18 and 23 findings, conditioning improves three-seed macro AUROC from 0.7687 to 0.8268 ($\Delta = +0.0581$, patient-bootstrap 95% CI [0.0482, 0.0685], $p < 0.001$) and macro AUPRC from 0.3592 to 0.4616, with a positive gain in every prespecified age bin, both sexes, both examination-year strata, and all four prespecified high-volume acquisition-site strata. A historical age ≥ 5 validation cohort of 1,507 studies replicates the effect (0.7933 ± 0.0017 to 0.8396 ± 0.0015), and blanking or shuffling indications there returns AUROC to 0.8092 and 0.7960, so the gain depends on the *correct* patient context. A no-CT control shows that indication text alone already reaches 0.7860. The multimodal model exceeds both it and the matched CT-only model. Finally, a locally measured benchmark of ten methods on the same pediatric cohort shows that released adult CT foundation models transfer poorly to children. A 2-D biomedical encoder with a linear probe outperforms every volumetric CT model we could reproduce. This underscores why pediatric performance must be measured rather than inherited from adult tables.

1 Introduction

Volumetric chest CT vision–language systems such as CT-CLIP [1], MPS-CT [3], GreenRFM [4], and ARC-CT [2] have raised adult abnormality-classification performance rapidly, largely on the CT-RATE benchmark. Two properties of that progress do not carry into pediatric practice. First, the models are trained on adults and evaluated on

adults. Pediatric anatomy is smaller, disease prevalence is different, and thin-slice reconstruction geometry differs. No public pediatric chest-CT benchmark exists against which transfer can be checked. Second, the standard read-out is a positive/negative pathology-prompt softmax that ignores the clinical question. In pediatric radiology that question is seldom generic: indications encode malignancy history, transplant status, postoperative context, cystic fibrosis surveillance, or a focused diagnostic concern, and they alter disease priors substantially before a single voxel is examined.

A naive remedy is to concatenate a global metadata embedding to the image embedding. That remedy imposes the same correction on unrelated pathologies, so helpful and harmful effects can cancel. We instead condition pathology-specific query slots on clinical context while preserving a separate, metadata-free general representation, so that the model can benefit from the indication for the pathologies it concerns and leave the rest untouched.

This paper makes four contributions specific to pediatric chest CT:

- a metadata-conditioned Q-Former that combines pathology-specific clinical cross-attention, identity-initialized FiLM, and a class-specific clinical-relevance gate, built on an anatomy-routed 3-D encoder.
- a frozen, MRN-disjoint pediatric evaluation protocol covering ages 0–18 with 23 pediatric-relevant findings, three seeds per arm, patient-bootstrap confidence intervals, and prespecified age, sex, examination-year, and site subgroups. It is replicated on a second historical pediatric cohort.
- causal and leakage controls that separate genuine visual–clinical fusion from text-only shortcuts. The controls are correct versus blank versus shuffled indications, no-CT metadata-only heads, and stratification by explicit target terminology in the indication.
- the first locally measured pediatric benchmark of released chest-CT foundation models, showing how far adult rankings fail to transfer and which model families degrade most.

2 Method

Figure 1 gives an overview of the model. The following subsections define each component.

2.1 Base volumetric representation

Given a CT volume X , an R(2+1)D-18 encoder produces a 512-channel spatial feature map. The encoder starts from Kinetics weights and is first adapted for eight epochs on the 23 pediatric labels with asymmetric multilabel loss. The checkpoint with the highest development macro AU-ROC initializes Stage 2. Thus, the primary < 18 experiment follows the ARC-CT anatomy-query design but does *not* load an adult ARC-CT checkpoint. Its transferable priors are the Kinetics visual initialization and pretrained biomedical language encoders.

Stage 2 uses a four-layer, eight-head, 768-dimensional anatomy-aware Q-Former. For C pathology targets, the general bank contains 10 anatomy queries, C pathology queries, and two global queries. Its pooled representation z_{gen} is deliberately isolated from metadata-conditioned slots, retaining a canonical metadata-free image/report embedding. The 10 pediatric regions are the five lung lobes, central airway and esophagus, heart and great vessels, pleural space, chest wall and skeleton, and upper abdomen. All 6,341 studies in the primary splits have a matched TotalSegmentator-derived mask. The primary recipe requires that mask and restricts each anatomy query to its assigned region. Mask-free operation is a registered ablation rather than a silent fallback.

2.2 Clinical context encoder

The clinical context combines indication text I , a learned age representation a , sex s , and an age–sex interaction:

$$H_C = \text{ContextEncoder}(I, a, s). \quad (1)$$

The indication is tokenized to at most 64 tokens and encoded as a *token sequence*, rather than a single pooled vector, by a separate frozen CXR-BERT. Three learned 768-dimensional demographic tokens are appended: age, sex, and their interaction. Age first enters a 64-dimensional ordinal embedding, sex a 16-dimensional embedding with female, male, and unknown rows, and their interaction a 32-dimensional table. Separate linear projections map all three to the Q-Former width. The interaction table and its projection start at zero so that an age–sex interaction is learned only when supported by data.

The age representation is an ordered cumulative embedding,

$$e_{\text{age}}(b) = e_0 + \sum_{j < b} \text{softplus}(\delta_j), \quad (2)$$

which enforces band order while allowing the distance between adjacent bands to be learned. The complete shared vocabulary contains 10 half-open bands: $[0, 1)$, $[1, 2)$, $[2, 5)$, $[5, 10)$, $[10, 13)$, $[13, 18)$, $[18, 25)$, $[25, 40)$, $[40, 60)$, and $[60, \infty)$ years, plus a separate unknown row. Only the first six occur in the primary pediatric cohort. This is distinct from the five wider age bins used only for subgroup reporting.

Missing or unusable indications are represented by the single canonical text “no clinical indication provided.” Age

and sex have explicit unknown rows and are never imputed from other patients. The primary arm randomly replaces 30% of training indications with the canonical missing string, while retaining age and sex, to prevent brittle dependence on text. No indication dropout is applied at evaluation. The historical replication checkpoints used scalar age and 10% indication dropout.

2.3 C1: pathology-specific conditioning

Each general pathology query has a conditioned twin. The conditioned query attends to H_C through a gated cross-attention residual and is then modulated by FiLM:

$$\tilde{q}_c = q_c + g_c \text{CrossAttn}(q_c, H_C), \quad (3)$$

$$q_c^{\text{ind}} = \gamma_c(H_C) \odot \tilde{q}_c + \beta_c(H_C). \quad (4)$$

The modulation is initialized near identity ($\gamma \approx 1$, $\beta \approx 0$), so the pretrained CT pathway remains stable early in adaptation. A clinical/global conditioned slot is added alongside the C pathology twins, giving $13 + 2C = 59$ slots for the 23-class pediatric task.

2.4 C2: class-specific clinical relevance

For each pathology c , C2 compares the indication embedding e_I with a frozen class-prompt embedding p_c :

$$r_c = \sigma(\text{MLP}[e_I; p_c; e_I \odot p_c]), \quad (5)$$

$$w_c = 1 + \text{softplus}(b)r_c. \quad (6)$$

The non-negative weight w_c controls conditioned-pathology pooling and the conditioned prompt loss. With $b = -2$ at initialization, C2 starts as a modest correction and can never push an incidental finding below its unconditioned weight.

2.5 Fusion and readouts

The primary model pools the C conditioned pathology twins with C2 weights and averages that representation with the clinical/global slot:

$$z_{\text{pool}} = \frac{\sum_c w_c t_c^{\text{ind}}}{\sum_c w_c}, \quad z_{\text{ind}} = \frac{1}{2}(z_{\text{pool}} + z_{\text{clin}}), \quad (7)$$

$$z_{\text{final}} = W_f[z_{\text{gen}}; z_{\text{ind}}], \quad W_f \leftarrow [I \ 0]. \quad (8)$$

Both branches use the same positive/negative pathology-prompt softmax. The general branch reads z_{gen} and the conditioned branch reads z_{final} . Because W_f starts as $[I \ 0]$ and every conditioning path is zero-gated or identity-initialized, the conditioned readout equals the general readout at step zero. Everything the conditioned branch learns is therefore a departure from the CT-only parent. A separate supervised `ctx_cls` head (LayerNorm plus one linear layer on $[z_{\text{gen}}; z_{\text{ind}}]$) is evaluated only as a labeled ablation and never conflated with prompt inference.

Table 1: Illustrative model inputs. Indications are paraphrased composites, not verbatim patient records. They demonstrate the clinical information available to the model without disclosing protected health information.

Age band	Sex token	Paraphrased indication	Information conveyed	Candidate affected findings
[0, 1)	female	Postoperative congenital heart disease; new respiratory distress	Surgical context and acute change	medical material, atelectasis, effusion
[2, 5)	male	Treated solid tumour; surveillance for pulmonary spread	Oncologic history and purpose of examination	pulmonary metastases, nodules
[5, 10)	female	Cystic-fibrosis follow-up; assess airway disease	Chronic diagnosis and targeted question	bronchiectasis, mucus plugging
[13, 18)	male	Fever after stem-cell transplantation; evaluate opportunistic infection	Immunosuppression and acute symptom	nodules, opacity, consolidation
unknown	unknown	no clinical indication provided	Explicit missingness, not fabricated text	no class-specific textual prior

2.6 Training objective

Stage 1 trains the Kinetics-initialized R(2+1)D-18 encoder and a temporary 23-output linear head with asymmetric loss for eight epochs (batch size 8, learning rate 10^{-4} , weight decay 10^{-4}). The head is discarded after development selection. Stage 2 loads the selected visual encoder and jointly optimizes the image pathway, Q-Former, visual/text projections, CXR-BERT LoRA parameters, and the new context modules. It uses batches of four with eight gradient-accumulation steps, at most 10,000 optimizer updates, a 500-update warm-up, and development evaluation every 500 updates. The base vision learning rate is 10^{-5} , the text learning rate 10^{-7} , and the zero-initialized context modules use a $20\times$ multiplier. Base and context gradient norms are clipped separately.

The Stage-2 pediatric recipe optimizes

$$\mathcal{L} = \mathcal{L}_{\text{clip}} + 0.5\mathcal{L}_{\text{prompt}} + 0.5\mathcal{L}_{\text{align}} + 0.5\mathcal{L}_{\text{ptok}} + 2.0\mathcal{L}_{\text{ind}}, \quad (9)$$

where $\mathcal{L}_{\text{clip}}$ aligns whole-volume and report embeddings, $\mathcal{L}_{\text{prompt}}$ is the general multilabel prompt loss, $\mathcal{L}_{\text{align}}$ aligns anatomy-pooled image features with organ-localized findings text, $\mathcal{L}_{\text{ptok}}$ supervises pathology tokens, and $\mathcal{L}_{\text{ind}}$ is the conditioned prompt loss. Training uses temperature one, null/empty negative prompts, no L2 normalization, and checkpoint selection by conditioned development AUROC. The counterfactual loss and auxiliary `ctx_cls` loss are zero in the primary arm and appear only in named ablations. No CT-RATE scan enters a pediatric minibatch, and no pediatric test study participates in optimization or checkpoint selection.

3 Pediatric Cohort and Protocol

3.1 Source archive, eligibility, and privacy

The retrospective source archive contains 8,870 chest-CT report records from a single children’s hospital. Existing conversion and quality-control filters produced 8,760 model-ready thin-slice volumes. The present paper then retained the 6,341 examinations acquired at $0 \leq \text{age} < 18$.

The examinations span 2011–2026 and are non-contrast chest CTs. Every included study has a volume, report, 23-label vector, clinical-context row, and anatomy mask. No included study is silently dropped for missing imaging or segmentation.

Raw identifiers are used only inside the institutional computing environment to establish the MRN-level split. Model-facing tables are keyed by a de-identified volume identifier. The indication is parsed from the report, then scrubbed against row-specific patient identifiers, accession numbers, dates, a cohort-wide patient/provider name pool, and structural identifier patterns. Fatal pre-flight checks reject surviving MRNs, accessions, long digit strings, dates, contact information, duplicate keys, or incomplete split coverage. Report and indication processing is performed locally. Clinical text is not sent to an external service.

3.2 Report-derived label construction

Findings and impression sections are divided into numbered sentences and processed locally by Qwen3-8B under constrained JSON decoding. For each of 27 candidate findings, the extractor returns a binary decision and the index of the supporting report sentence. Out-of-range indices and malformed records are detectable failures rather than silent all-zero labels. The paper task removes four adult-specific or inadequately supported targets from both training and evaluation, leaving 23 outputs. The removed targets are arterial wall calcification, coronary artery calcification, hiatal hernia, and emphysema. The labels are therefore report-derived weak supervision, not independent image adjudications.

The full findings text is encoded during Stage 2 for global image-report alignment. Organ-specific text for the primary run is selected from the findings section by anatomy-keyword matching. The available Qwen organ-sentence cache is a named ablation and is not used in the primary result. Reports are training supervision only: at inference the model receives the CT, scrubbed indication, age band, sex, and fixed pathology prompts, but never the report whose findings define the reference labels.

Figure 1: Architecture for the 23-finding pediatric task. The metadata-independent general bank is retained. Pathology-specific conditioned twins attend to clinical context through C1 cross-attention and FiLM and are pooled by C2 relevance weights. Prompt inference is the primary readout. `ctx_cls` is a separate supervised ablation.

Figure 2: Assembly of the primary pediatric cohort. Train, development, and test are mutually exclusive at the MRN/patient level. The historical source validation pool is retained as the frozen test cohort. The source training pool alone is divided into train and development.

3.3 CT and anatomy preprocessing

Volumes are stored in Hounsfield units and deterministically padded or cropped to $192 \times 192 \times 96$ voxels, with out-of-field values clamped to the -1024 HU air floor before padding. Three channels encode lung, soft-tissue, and bone windows. Stage 1 uses a lung window of $[-1500, 500]$ HU and Stage 2 uses $[-1000, 400]$ HU. Both use $[-160, 240]$ and $[300, 2000]$ HU for soft tissue and bone. Training augmentation pads to $240 \times 240 \times 120$ before a paired random image/mask crop, permits anatomy-preserving flips, and applies HU intensity jitter (30 HU in Stage 1 and 20 HU in Stage 2). Evaluation uses the deterministic center geometry.

TotalSegmentator structures are merged into the 10 pediatric regions used by the anatomy queries. Distributed targets such as medical material, lymphadenopathy, post-treatment change, and mass/neoplasm remain unrestricted rather than being assigned a confidently wrong region. Image and mask undergo identical spatial transforms.

3.4 Two evaluation protocols

Table 2 summarizes the two pediatric protocols used in this paper.

Frozen < 18 protocol (primary). Every study with $0 \leq \text{age} < 18$ is retained, including ages 0–5. The historical training pool was divided by medical record number (MRN) into 4,320 training studies (2,108 patients) and 748 development studies (373 patients). The historical patient-disjoint validation pool contributes 1,273 test studies (611 patients). Patient overlap is zero for every pair of splits. Because this source test pool was inspected during earlier exploratory work, we describe it as a *frozen retrospective* test rather than a never-observed prospective one.

Historical age- ≥ 5 protocol (replication). An earlier campaign excluded examinations below age five and used 5,948 training and 1,509 validation studies with patient-disjoint splitting. An MRN-level audit confirmed no patient overlap between those pools. The same audit found that the 1,507 scored studies were contained in the 1,509-study Stage-2 model-selection validation list. We therefore label every result on this cohort as historical held-out *validation*, never as sealed test performance, and use it as a replication cohort for the frozen protocol.

Unfiltered all-age sensitivity cohort. To measure the joint effect of retaining both children younger than five and patients who have reached adulthood, we additionally use every study in the clean source split without

Table 2: Pediatric cohorts and matched experimental splits.

Protocol	Train	Development	Test / validation	Classes	Provenance
Frozen < 18 (primary)	4,320	748	1,273	23	MRN-disjoint train/development/test; includes ages 0–5; 2,108 / 373 / 611 patients
Historical age ≥ 5 (replication)	5,948	1,509	N/A	23	Patient-disjoint train/validation; the 1,507 scored studies (648 patients) are a subset of the model-selection validation pool

an age cutoff. This cohort spans 0.01–69 years and contains 7,001 training studies from 3,180 patients and 1,761 validation studies from 791 patients, with zero MRN or study overlap. Its Stage-1 encoder is retrained on the same all-age training pool, followed by three full-model and three matched CT-only Stage-2 seeds. Quantitative results will enter the paper only after all six paired evaluations complete.

The cohort is strongly multilabel: 90.7% of studies have at least one positive target, and the mean is 3.26 positives per study. Prevalence ranges from roughly 2% for pneumothorax, mass/neoplasm, and cardiomegaly to approximately 49% for lung nodules. Train, development, and test prevalences were checked during MRN-grouped split selection. Complete counts are reported in Table 20.

3.5 Evaluation

The primary endpoint is macro AUROC across the 23 targets. Macro AUPRC, Brier score, expected calibration error, and sensitivity at 95% specificity are secondary. Every matched CT-only and full-model comparison uses three independent seeds. Confidence intervals for the primary paired differences use 2,000 bootstrap replicates that resample *patients*, not studies, because the pediatric cohorts average more than two studies per child and study-level resampling understates the interval. Age strata include only classes with at least 10 positives and 10 negatives. For the local benchmark, non-AUROC metrics follow the CT-CLIP/ARC-CT convention.

3.6 Controls for context use and leakage

Three controls distinguish genuine visual-clinical fusion from text-only shortcuts.

- **Indication intervention.** Each trained checkpoint is evaluated without retraining with (i) the correct indication, (ii) all indications blanked to `NO_INDICATION`, and (iii) indications shuffled across patients. This isolates semantic context use from added capacity.
- **No-CT controls.** The same frozen indication encoder is fitted with only a LayerNorm and a linear 23-label head, using no image input, for indication alone, demographics alone, and their combination. A multimodal model must beat both the matched CT-only model and this control.

- **Terminology stratification.** A conservative case-insensitive audit flags studies whose indication explicitly names at least one positive target. Performance is reported separately for flagged and unflagged studies.

3.7 Registered component matrix

A 35-arm matrix changes one factor at a time around the full recipe: CT-only, C1 off, C2 off, cross-attention only, FiLM only, indication only, demographics only, no metadata, naive pooled-metadata fusion, anatomy queries off, routing off, regional alignment off, mask-free operation, scalar versus banded age, indication dropout 0/0.1/0.3/0.4, augmentation off, counterfactual loss, Qwen versus keyword region text, the auxiliary `ctx_cls` head, L2 normalization, standard versus null negative prompts, indication-loss weights 1/2/4, one/two/six global queries, and general-versus-conditioned checkpoint selection. We also register imbalance-focused loss ablations: asymmetric loss (ASL) and batch nuclear-norm maximization (BNM) added to BCE at weights 0.001, 0.003, and 0.010, as well as their combinations. For a batch probability matrix $P \in [0, 1]^{B \times C}$, BNM contributes $-\lambda \|P\|_* / B$ and therefore encourages diverse, higher-rank predictions; it does not rebalance labels directly. Cells are populated only from completed artifacts matched to their checkpoint, split, and readout. Unfinished arms remain in the experiment registry and are omitted from result tables.

4 Results on the Frozen < 18 Test Split

Status of these numbers. All values in this section are from a prespecified checkpoint snapshot of the ongoing frozen campaign. They are interim rather than converged, and the tables below regenerate automatically as arms complete. We report them because the primary effect is already large relative to its interval, not because they are final.

4.1 Primary effect

Across three seeds, metadata-conditioned prompt inference improves frozen-test macro AUROC from 0.7687 to

Table 3: Characteristics of the primary MRN-disjoint pediatric cohort. Word counts refer to present indications and to the de-identified findings section, respectively. Percentages use studies, not patients, as the denominator.

Characteristic	Train	Development	Frozen test	All
Studies	4,320	748	1,273	6,341
Patients	2,108	373	611	3,092
Age, mean $\pm$ SD (years)	10.19 $\pm$ 5.23	9.96 $\pm$ 5.31	10.17 $\pm$ 5.11	10.16 $\pm$ 5.22
Age, median [IQR] (years)	11 [6–15]	11 [5–15]	11 [6–15]	11 [6–15]
Female, n (%)	1,941 (44.93)	358 (47.86)	681 (53.50)	2,980 (47.00)
Male, n (%)	2,372 (54.91)	390 (52.14)	592 (46.50)	3,354 (52.89)
Unknown sex, n (%)	7 (0.16)	0	0	7 (0.11)
Indication present, n (%)	4,303 (99.61)	746 (99.73)	1,269 (99.69)	6,318 (99.64)
Indication words, median [IQR]	22 [16–31]	22 [16–32]	22 [16–31]	22 [16–31]
Findings words, median [IQR]	156 [114–214]	147 [110–208]	158 [117–212]	156 [114–213]
Zero positive targets, n (%)	418 (9.68)	52 (6.95)	120 (9.43)	590 (9.30)
Positive targets/study, mean (median)	3.26 (3)	3.23 (3)	3.25 (3)	3.26 (3)
Matched CT and anatomy mask, n (%)	4,320 (100)	748 (100)	1,273 (100)	6,341 (100)

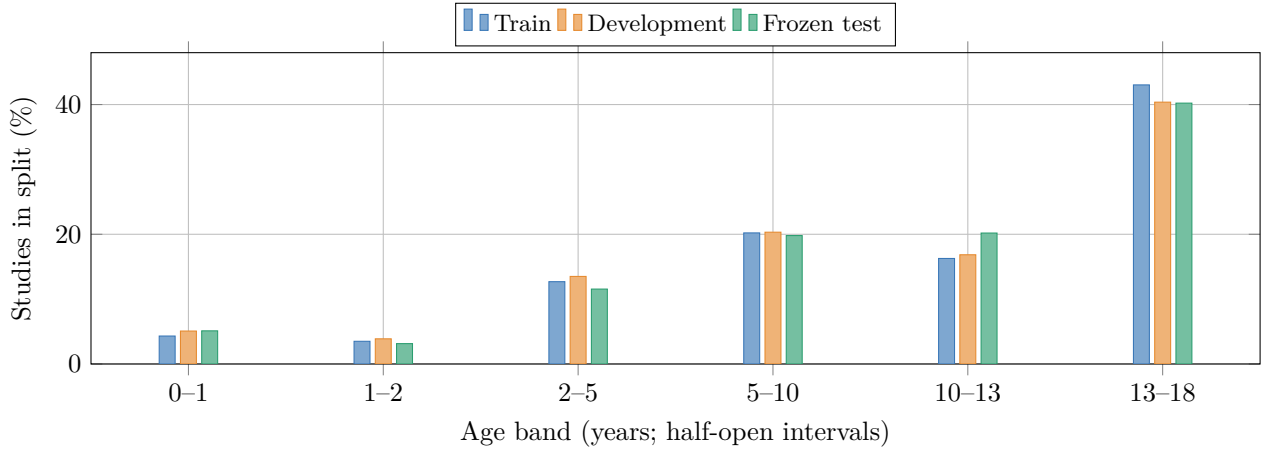


Figure 3: Age distribution in the primary cohort. Every model age band appears in every split. No age is missing. The subgroup analysis instead merges these into 0–< 2, 2–< 5, 5–< 10, 10–< 15, and 15–< 18 bins to provide more stable class counts.

Table 4: Interim 20:00 checkpoint snapshot on the MRN-disjoint < 18 test split; training continued after evaluation.

Seed	CT-only AUROC	Full AUROC	Δ
0	0.7719	0.8173	+0.0454
1	0.7687	0.8313	+0.0625
2	0.7654	0.8317	+0.0662
Mean	0.7687	0.8268	+0.0581

Patient-bootstrap 95% CI for the mean paired AUROC difference: [0.0482, 0.0685]; two-sided paired bootstrap $p < 0.001$.

0.8268 ($\Delta = +0.0581$, patient-bootstrap 95% CI [0.0482, 0.0685], two-sided paired bootstrap $p < 0.001$, Table 4) and macro AUPRC from 0.3592 to 0.4616 ($\Delta = +0.1024$, 95% CI [0.0830, 0.1198]). Every seed improves by at least 0.045 AUROC. Sensitivity at 95% specificity rises from roughly 0.33 to 0.43–0.45 across seeds (Table 5). Calibration is not uniformly better: the full model’s Brier score and ECE are higher than CT-only’s for two of three seeds,

so the conditioned branch is more discriminative but not yet better calibrated, a point we return to in the limitations.

4.2 Consistency across subgroups

The AUROC gain is positive in every prespecified stratum (Table 6): all five age bins from 0–2 through 15–18 years, both sexes, both examination-year strata, and all four prespecified high-volume acquisition-site strata. The smallest absolute gain is in the 0–2-year bin (0.7185 to 0.7609), where indications are least specific and sample size is smallest. The largest is in the 10–15-year bin (0.7825 to 0.8478). That the gain survives at the youngest ages, where CT-only performance is weakest, argues against the conditioned branch merely amplifying an already-easy subpopulation.

Table 5: Audited completed < 18 checkpoint-snapshot and follow-up ablations. Arms without a completed frozen-test evaluation are omitted.

Arm	AUROC	AUPRC	Brier	ECE	Sens.@95% spec.
Full model, seed 0	0.8173	0.4619	0.1155	0.1093	0.4307
Full model, seed 1	0.8313	0.4547	0.1430	0.1546	0.4316
Full model, seed 2	0.8317	0.4681	0.1069	0.0941	0.4504
CT-only, seed 0	0.7719	0.3547	0.1012	0.0794	0.3321
CT-only, seed 1	0.7687	0.3681	0.0959	0.0637	0.3303
CT-only, seed 2	0.7654	0.3547	0.1025	0.0802	0.3263
C2 off	0.8126	0.4191	0.1119	0.1059	0.3842
C1 off	0.7629	0.3669	0.1292	0.1137	0.3534
R3D-18 backbone, full model, seed 0	0.8144	0.4549	0.1010	0.0879	0.4227
Asymmetric loss (ASL), seed 0	0.8055	0.4493	0.1016	0.0991	0.4017
BCE + BNM ($\lambda = 0.001$), seed 0	0.8305	0.4715	0.0989	0.0880	0.4430
BCE + BNM ($\lambda = 0.003$), seed 0	0.8301	0.4693	0.1000	0.0860	0.4370
BCE + BNM ($\lambda = 0.010$), seed 0	0.8298	0.4676	0.0955	0.0813	0.4292
ASL + BNM ($\lambda = 0.001$), seed 0	0.8190	0.4558	0.1882	0.2591	0.4118
ASL + BNM ($\lambda = 0.003$), seed 0	0.8172	0.4528	0.1614	0.2271	0.3998
ASL + BNM ($\lambda = 0.010$), seed 0	0.8158	0.4462	0.2103	0.2832	0.4110

Table 6: Age and sex subgroup consistency on the < 18 test split.

Subgroup	Model	N	AUROC
age_0_2	ct_only	105	0.7185
age_0_2	full	105	0.7609
age_2_5	ct_only	147	0.7509
age_2_5	full	147	0.8051
age_5_10	ct_only	252	0.7200
age_5_10	full	252	0.7807
age_10_15	ct_only	435	0.7825
age_10_15	full	435	0.8478
age_15_18	ct_only	334	0.7427
age_15_18	full	334	0.8066
sex_F	ct_only	681	0.7715
sex_F	full	681	0.8315
sex_M	ct_only	592	0.7603
sex_M	full	592	0.8179
exam_year_2011_2018	ct_only	615	0.7809
exam_year_2011_2018	full	615	0.8427
exam_year_2019_2026	ct_only	658	0.7587
exam_year_2019_2026	full	658	0.8090
acquisition_site_rank_1	ct_only	413	0.7516
acquisition_site_rank_1	full	413	0.8057
acquisition_site_rank_2	ct_only	293	0.7089
acquisition_site_rank_2	full	293	0.7816
acquisition_site_rank_3	ct_only	135	0.7164
acquisition_site_rank_3	full	135	0.7792
acquisition_site_rank_4	ct_only	116	0.7078
acquisition_site_rank_4	full	116	0.7649

4.3 Does excluding young children help?

A recurring practical question is whether ages 0–5, with their distinct anatomy and clinical spectrum, should be excluded from training. The age-training factorial answers it in the negative for the full model. On the identical 5– < 18 test subset, training only on ages 5– < 18 changes the full model’s AUROC by -0.0075 (95% CI $[-0.0136,$

Table 7: Performance stratified by explicit target terminology in the indication.

Terminology	Model	AUROC	AUPRC
explicit_term_present	ct_only	0.6694	0.6771
explicit_term_present	full	0.7153	0.7339
explicit_term_absent	ct_only	0.7692	0.3494
explicit_term_absent	full	0.8249	0.4486

-0.0008]) but changes the CT-only model’s by $+0.0250$ (95% CI $[0.0154, 0.0344]$, Table 9). The two models therefore respond in opposite directions: a CT-only encoder benefits from a narrower age range, whereas a model that can read the indication extracts useful signal from the younger children and is slightly harmed by their removal. We keep ages 0–5 in the primary cohort.

4.4 Which component carries the gain?

In the single-seed component snapshot, removing C1 lowers AUROC from 0.8173 to 0.7629, below the matched seed-0 CT-only value of 0.7719. Removing C2 yields 0.8126 (Table 5). Token-level C1 conditioning is therefore the larger contributor at this checkpoint. The converged matrix is required before a final component claim. Replacing the R(2+1)D-18 backbone with R3D-18 changes the matched seed-0 AUROC from 0.8173 to 0.8144, indicating little backbone sensitivity. A completed follow-up that replaces the Stage-2 classification term with asymmetric loss reaches 0.8055 AUROC and 0.4493 AUPRC on the same frozen test split. It does not exceed the 0.8268 full-model snapshot and is therefore not selected as the primary recipe. In contrast, adding BNM to BCE gives 0.8305, 0.8301, and 0.8298 AUROC for $\lambda = 0.001, 0.003,$ and 0.010 , respectively, with corresponding AUPRC of 0.4715, 0.4693, and 0.4676. The best setting improves

Table 8: Complete three-seed per-class results on the frozen < 18 test split.

Class	Pos.	CT AUC	Full AUC	Δ AUC	CT AUPRC	Full AUPRC	Δ AUPRC
Medical material	535	0.8941	0.9225	+0.0283	0.8649	0.9061	+0.0412
Cardiomegaly	29	0.9185	0.9609	+0.0424	0.2857	0.4413	+0.1556
Pericardial effusion	44	0.8471	0.9024	+0.0553	0.1890	0.2753	+0.0862
Lymphadenopathy	96	0.7745	0.7602	-0.0142	0.2616	0.2732	+0.0115
Atelectasis	447	0.7270	0.7465	+0.0195	0.6210	0.6560	+0.0350
Lung nodule	631	0.6579	0.6891	+0.0312	0.6223	0.6656	+0.0433
Lung opacity	426	0.7438	0.7741	+0.0303	0.6080	0.6536	+0.0456
Pulmonary fibrotic sequela	235	0.6614	0.7682	+0.1068	0.3138	0.4332	+0.1194
Pleural effusion	48	0.9559	0.9640	+0.0081	0.5193	0.6318	+0.1125
Mosaic attenuation pattern	185	0.7351	0.8253	+0.0901	0.3540	0.4362	+0.0823
Peribronchial thickening	168	0.7960	0.8352	+0.0392	0.4607	0.5047	+0.0439
Consolidation	76	0.8632	0.8922	+0.0290	0.4035	0.4380	+0.0346
Bronchiectasis	103	0.9004	0.9337	+0.0334	0.5713	0.7101	+0.1388
Interlobular septal thickening	76	0.8444	0.8818	+0.0374	0.3224	0.4314	+0.1090
Post-surgical or post-treatment change	392	0.6754	0.8150	+0.1396	0.4952	0.6828	+0.1876
Pulmonary metastases	76	0.6155	0.8170	+0.2014	0.1397	0.2708	+0.1311
Tree-in-bud	90	0.7893	0.8602	+0.0709	0.3177	0.4501	+0.1324
Pulmonary cyst	50	0.6645	0.7251	+0.0606	0.0690	0.1199	+0.0509
Mass or neoplasm	33	0.7224	0.8218	+0.0994	0.1285	0.2556	+0.1271
Mucus plugging	44	0.7891	0.8378	+0.0487	0.1760	0.2986	+0.1226
Pleural thickening or nodule	146	0.6757	0.6950	+0.0193	0.2283	0.2630	+0.0347
Bone lesion or fracture	179	0.5925	0.6699	+0.0774	0.2016	0.3056	+0.1040
Pneumothorax	25	0.8363	0.9176	+0.0814	0.1072	0.5136	+0.4065

Table 9: Interim 20:00 age-training checkpoint snapshot evaluated on the identical $5-< 18$ MRN-disjoint test sub-set.

Model	Training ages	AUROC	AUPRC
CT-only	$0-< 18$	0.7656	0.3573
CT-only	$5-< 18$	0.7907	0.4110
Full	$0-< 18$	0.8287	0.4689
Full	$5-< 18$	0.8212	0.4503

CT-only: excluding ages $0-5$ changes AUROC by $+0.0250$ (patient-bootstrap 95% CI $[0.0154, 0.0344]$).

Full: excluding ages $0-5$ changes AUROC by -0.0075 (patient-bootstrap 95% CI $[-0.0136, -0.0008]$).

over the matched seed-0 full-model snapshot by 0.0132 AUROC and also improves its Brier score (0.0989 versus 0.1155) and ECE (0.0880 versus 0.1093). Because these BNM results currently use one seed and the comparator is a checkpoint snapshot, we treat the finding as an exploratory loss ablation rather than changing the three-seed primary claim. Combining ASL with BNM does not improve further: the three weights reach 0.8190, 0.8172, and 0.8158 AUROC. Although these values recover part of ASL’s loss relative to BCE+BNM, all remain below BCE+BNM and their ECE ranges from 0.2271 to 0.2832. Thus ASL and BNM are not complementary under the present recipe, and BCE+BNM at $\lambda = 0.001$ is the loss variant carried forward for replication.

4.5 Qualitative localization

Figure 4 compares gradient-based class-localization maps from the metadata-conditioned model and its matched CT-only control on one MRN-disjoint pediatric test study. The four displayed targets are positive in the study-level ground truth. Each panel uses the peak axial slice for that model-target score. Both rows retain their normal anatomy-routed inference, and the routing masks themselves are not overlaid. This visualization is qualitative and does not enter checkpoint selection or quantitative evaluation.

5 Replication on the Historical Age- ≥ 5 Cohort

5.1 Matched three-seed comparison

On the historical 1,507-study validation cohort, the metadata-conditioned model improves three-seed macro AUROC from 0.7933 ± 0.0017 to 0.8396 ± 0.0015 ($+0.0463$, Table 10). All three conditioned seeds exceed 0.83 and each improves over its own seed-matched control (Table 11). The improvement is positive for all 23 pathologies (Table 19), with the largest gains for pneumothorax ($+0.164$), pulmonary metastases ($+0.137$), post-surgical or post-treatment change ($+0.117$), pulmonary cysts ($+0.103$), and mosaic attenuation ($+0.096$). These are the findings whose prior probability is most strongly shaped by the clinical history a pediatric indication typically carries.

Table 10: Historical age- ≥ 5 matched prompt-inference validation results (macro AUROC, mean $\pm$ SD over three seeds, $N = 1,507$). These are not independent test estimates.

Cohort	CT-only	METAQ-CT	Δ
Age ≥ 5 , 23 labels	0.7933 $\pm$.0017	0.8396$\pm$.0015	+0.0463

Table 11: Historical cohort: seeds and inference-time indication intervention on the same trained checkpoints.

Arm	Seed 0	Seed 1	Seed 2	Mean
CT-only	0.7934	0.7950	0.7916	0.7933
METAQ-CT, correct indication	0.8386	0.8414	0.8389	0.8396
METAQ-CT, indication blanked	0.8111	0.8091	0.8074	0.8092
METAQ-CT, indication shuffled	0.7980	0.7967	0.7934	0.7960

Table 12: Historical cohort performance by age. Only eligible classes enter each stratum’s macro AUROC.

Age	N	Classes	CT-only	METAQ-CT (Δ)
5–10	252	15	0.7451	0.7904 (+.0452)
10–18	768	23	0.7995	0.8440 (+.0445)
18–25	402	20	0.7729	0.8190 (+.0461)
25+	85	15	0.7615	0.8050 (+.0435)
All < 18	1,020	23	0.7979	0.8412 (+.0433)
All ≥ 18	487	23	0.7841	0.8325 (+.0483)
Pooled	1,507	23	0.7933	0.8396 (+.0463)

The current pediatric Stage-1 and banded-age recipe independently reproduces this result on the same age- ≥ 5 cohort. Its three conditioned seeds score 0.8370, 0.8393, and 0.8399 (mean 0.8387 ± 0.0015), compared with 0.8177, 0.8159, and 0.7813 for matched CT-only training (mean 0.8050 ± 0.0205). The paired mean gain is therefore +0.0338. The near-identical absolute AUROC of the historical Refined-C1 recipe (0.8396) and current recipe (0.8387), despite different initialization and demographic encoding, supports replication rather than dependence on one training implementation.

5.2 Consistency across age

The gain is also positive in every age stratum of the historical cohort (Table 12), including the 5–10-year bin where CT-only performance is lowest (0.7451 to 0.7904). The historical cohort extends above 18 years because the source institution images young adults with childhood-onset disease. Gains in those strata are of the same size as in children.

5.3 Completed historical age- ≥ 5 current-recipe ablations

Every current-recipe arm with a completed evaluation on the historical validation split is listed in Table 13. Unfinished arms are kept in the experiment registry rather than

displayed as empty paper rows.

6 Does the Model Use the Correct Indication?

6.1 Inference-time intervention

On the historical cohort, correct indications outperform blank and shuffled indications by 3.04 and 4.36 AUROC points respectively (Table 11, Fig. 5). Shuffling preserves the marginal distribution of indication text and destroys its pairing with the patient. It removes almost the entire gain over CT-only. The model is therefore not benefiting from extra capacity or from the mere presence of clinical language. It is reading the specific patient’s indication. The corresponding three-seed intervention on the frozen < 18 split is queued and will be reported in the same form.

6.2 How predictive is metadata without the CT?

Table 15 reports no-CT controls on the frozen < 18 test split. Age and sex alone are near chance (0.5064). Indication text alone, however, reaches 0.7860 macro AUROC, *higher than the matched CT-only model’s 0.7687*. Adding demographics to the indication adds only 0.0007. Two consequences follow. First, on this cohort the clinical text is a stronger single predictor than the image, which is itself a finding about pediatric indications. Second, beating CT-only is not sufficient evidence of multimodal fusion. The model must beat the text-only control as well. It does: the full model’s 0.8268 exceeds the indication-only head by 0.0408 and the CT-only model by 0.0581, i.e., it uses both sources.

6.3 Explicit terminology in the indication

A conservative audit found that at least one positive target was explicitly named in the indication for 1,064/4,320 (24.6%) training, 169/748 (22.6%) development, and 330/1,273 (25.9%) test studies. This heuristic cannot distinguish a suspected from a confirmed diagnosis, so it is not by itself evidence of leakage. Stratifying on it is informative nonetheless (Table 7): the conditioned model improves over CT-only both when the target is explicitly named (0.6694 to 0.7153) and when it is not (0.7692 to 0.8249). The gain in the *unflagged* majority is larger than in the flagged minority, which is the opposite of what a lexical shortcut would produce. Notably, absolute AUROC is lower in the flagged stratum for both models: studies whose indication already names a finding are harder, not easier, presumably because they are enriched for ambiguous follow-up examinations.

Table 13: Completed current-recipe age- ≥ 5 23-target ablations on the same historical validation split. Global denotes the conditioned/fused prompt readout; arms without a completed evaluation are omitted.

Arm	Global AUC	General AUC	Global F1	Global accuracy	Global precision
Full model, three-seed mean	0.8387	0.8143	0.3644	0.8675	0.5082
CT-only, three-seed mean	0.8050	0.8050	0.3303	0.8729	0.5174
Full current model, seed 0	0.8370	0.8076	0.3134	0.8695	0.5334
Full current model, seed 1	0.8393	0.8167	0.3400	0.8745	0.4924
Full current model, seed 2	0.8399	0.8185	0.4397	0.8586	0.4987
CT-only, seed 0	0.8177	0.8177	0.3738	0.8776	0.5820
CT-only, seed 1	0.8159	0.8159	0.3514	0.8792	0.5517
CT-only, seed 2	0.7813	0.7813	0.2657	0.8617	0.4185
Indication dropout 0.4	0.8216	0.7882	0.3314	0.8512	0.4987
C1 off	0.8103	0.7893	0.2896	0.8016	0.3982
Indication conditioning only	0.8204	0.7883	0.3591	0.8152	0.3997
Demographic conditioning only	0.8085	0.7890	0.3874	0.8298	0.4060
Standard negative prompts	0.8201	0.7767	0.3348	0.8257	0.4523
Auxiliary ctx_cls	0.8243	0.7880	0.3781	0.8391	0.4762
Indication-loss weight 1	0.8200	0.7862	0.3520	0.8173	0.4350
Indication-loss weight 4	0.8229	0.7893	0.3634	0.8359	0.4199
Qwen region text	0.8206	0.7877	0.4036	0.7958	0.3817
One global query	0.8221	0.7879	0.3274	0.8478	0.3971
Six global queries	0.8238	0.7940	0.3294	0.8044	0.4276
Augmentation off	0.8223	0.7795	0.3587	0.8039	0.3523

Table 14: Available current-recipe indication interventions on the age- ≥ 5 cohort. Only completed evaluations are shown.

Seed	Indication	AUROC
0	Correct	0.8370
0	Blank	0.8180
1	Correct	0.8393
1	Blank	0.8204
1	Shuffled	0.8107
2	Correct	0.8399

Table 15: No-CT controls on the frozen < 18 test split ($N = 1,273$). AUROC is mean $\pm$ SD over three linear-head seeds. AUPRC is the three-seed mean.

Input	Macro AUROC	Macro AUPRC
Age + sex (no CT)	0.5064 $\pm$ 0.0065	0.1578
Indication (no CT)	0.7860 $\pm$ 0.0003	0.3792
Indication + age + sex (no CT)	0.7867 $\pm$ 0.0010	0.3801
CT-only (image only)	0.7687	0.3592
METAQ-CT (image + metadata)	0.8268	0.4616

7 Pediatric Benchmark of Released Chest-CT Models

7.1 Why a local benchmark is necessary

No published pediatric chest-CT benchmark exists, and the adult CT-RATE leaderboard cannot be assumed to transfer. We therefore evaluated every reproducible comparison method on the historical 1,507-study cohort with the identical 23-label file, the identical metric code, and

patient-grouped bootstrap intervals (Table 16). Methods were assigned their own published readout: GreenRFM’s supervised head, MPS-CT’s and CT-CLIP’s prompt softmax, and so on. For methods with multiple saved checkpoints, selection followed the same best-of- N validation rule our own models use. This changed both affected base-lines by at most 0.001 relative to their final checkpoints. Four further methods reported on CT-RATE could not be measured because their releases lack code, weights, a training entry point, or a license. Those methods are BIUD, ViSD-Boost, BrgSA, and fVLM. We say so rather than importing their adult numbers into a pediatric table.

7.2 What the benchmark shows

Three observations follow from Table 16.

Released adult models transfer poorly to children.

CT-CLIP and its VocabFine variant fall to 0.5478 and 0.5339, barely above chance, and H-LIP to 0.6916, from published adult AUROCs of 0.731 and 0.787. For CT-CLIP this is not a readout artifact: sweeping all four prompt templates shipped in its repository spans only 0.528–0.548, its image latents are not degenerate (103 of 512 dimensions carry 90% of variance), and a supervised probe fitted on those latents reaches only 0.609. The limitation is representational. A plausible mechanism is CT-CLIP’s fixed 360 mm in-plane window: a small child occupies a much smaller fraction of that frame than the adults it was trained on, and 29.5% of our pediatric reconstructions exceed the window and are cropped.

Pediatric adaptation helps every model, but unevenly. Adapting GreenRFM to the pediatric training split lifts it to 0.7477 and MPS-CT to 0.6961. Both remain 9–14 points below METAQ-CT and 5–10 points below our own CT-only control, which shares neither their architecture nor their readout but does share the anatomy-routed initialization.

A 2-D encoder with a linear probe beats every volumetric model. A linear probe on frozen BiomedCLIP features reaches 0.7643, exceeding GreenRFM, MPS-CT, and Merlin. Those features are 32 axial slices, mean-pooled, with no three-dimensional context at all. Its per-class profile explains why: it is competitive on findings legible in a single section and at chance (0.503) on lung nodules. We do not read this as evidence against volumetric modeling. Our own volumetric CT-only model at 0.7933 and full model at 0.8396 are well above it. We read it as evidence that current volumetric *pretraining*, as released, buys less on pediatric CT than its adult rankings suggest, and that pediatric performance must be measured rather than inherited.

8 Pediatric Report–CT Retrieval

8.1 Protocol

Classification asks whether a model can name a finding. Retrieval asks whether its image and report embeddings agree on *which study* is which. We therefore evaluated report→image and image→image retrieval on the same 1,507-study cohort, with no retrieval-specific fitting (Table 17). Report→image is exact-match Recall@K: each study’s report queries all 1,507 volumes and only its own counts as a hit, so chance is K/N . Every model encodes the byte-identical report string through its own text tower at its own trained context length. That string is the impression-first concatenation our own evaluator uses. Image→image follows the CT-CLIP/MPS-CT convention of mean label-set Jaccard between a query and its top- K retrieved studies, with one correction and one disclosure. The correction: the query is excluded from its own gallery. The disclosure: the evaluation script that prior work inherits from CT-CLIP does *not* exclude it, and also accumulates its running mean across successive K without resetting, so the values those papers label “MAP@K” are inflated by a guaranteed self-hit at $K=1$. We reproduced ARC-CT’s published 71.3/61.7/54.3 to within 0.3 points from our own CT-RATE latents by applying exactly that script. Table 17 therefore reports the clean metric as its headline and the inherited convention beside it, labelled, so the pediatric numbers remain comparable to existing adult tables without repeating their error.

8.2 Findings

Two structural facts shape the table. First, Refined C1 freezes the inherited base and trains only the conditioned

bank, so its metadata-free z_{gen} is identical to the seed-matched CT-only model’s, verified to floating-point noise on the stored latents. The metadata-free row is therefore shared. The pediatric-specific question is whether conditioning on indication, age, and sex changes retrieval. It is answered by the conditioned readouts, which use each gallery study’s own metadata and are labelled as not-equal-input. Second, report→image retrieval on children lands where it does on adults: our z_{gen} reaches R@5/10/50/100 of 10.8/17.1/41.2/54.2, against ARC-CT’s published adult CT-RATE 11.4/18.0/40.4/52.1, so the image–report alignment learned from pediatric reports is not weaker than the adult one.

The complete model sweeps the comparison set. METAQ-CT Refined C1 reading out $\text{norm}(z_{\text{gen}} + z_{\text{ind}})$ leads every prior method on all ten columns of Table 17: 9.5/15.5/39.9/51.7 on report→image against the best baseline’s 6.3/10.7/29.1/42.6, and 32.2/30.9/27.7 on clean image→image Jaccard against 30.9/28.4/24.2. The cross-model table carries only the full model and its matched CT-only control, so it compares methods rather than our own readouts. Table 18 separates the readouts. They split along an interpretable axis rather than dominating each other. The metadata-free z_{gen} retrieves the *exact* report best (10.8 at $K=5$), while the conditioned z_{ind} retrieves the most *label-similar* studies best (33.4 at $K=5$, and highest at every K under both conventions). Indication text sharpens what a study is a case *of* and blurs which particular report was dictated for it. That is the behaviour expected if conditioning injects clinical context rather than report-specific wording. The mixed readout sits between the two by construction.

Among comparison methods, MPS-CT adapted to the pediatric cohort is a genuine second (6.3/10.7/29.1/42.6). Released CT-CLIP and its VocabFine variant are at chance (R@5 of 0.86 and 0.33 against $K/N=0.33$), the retrieval counterpart of their near-chance classification transfer. GreenRFM is the instructive case: even after pediatric adaptation its report→image recall is at chance (0.33/1.13/5.37/9.42), while its image→image Jaccard (27.8 at $K=5$) shows the image space itself is informative. This is the same mechanism that invalidated its prompt readout. A contrastive objective on unnormalized dot products at fixed temperature does not learn a space in which cosine similarity between a report and its volume is meaningful. That is also why GreenRFM’s own published results rest on its supervised head. Of the released-weight rows, H-LIP is the strongest (3.6/5.8/14.9/22.4), above locally adapted GreenRFM but below MPS-CT and every arm of ours. BiomedCLIP, which sees the volume only as independent 2-D slices, reaches 1.9/3.3/11.8/17.4, and Merlin, an abdominal–pelvic model applied to chest CT, is nearest chance at 1.0/1.7/6.2/10.9 while retaining a usable image space (21.8 Jaccard at $K=5$).

One asymmetry is disclosed rather than corrected. Every model encodes the identical impression-first string. Each uses its own tokenizer at its own trained context length, and those lengths differ by an order of magnitude.

Table 16: Pediatric benchmark on the historical cohort (1,507 studies, 648 patients, 23 labels). All values are locally measured macro AUROC. Rows are grouped by whether the method was adapted to the pediatric cohort, because a pediatric-adapted model and an untrained transfer answer different questions. Intervals are 1,000-sample bootstraps resampling patients.

Method	Readout	AUROC	95% CI	Qualification
<i>Adapted to the pediatric cohort</i>				
METAQ-CT	conditioned prompt	0.8396^a	seed SD only	three seeds, ± 0.0015
Our CT-only	prompt	0.7933 ^a	seed SD only	matched three-seed control
BiomedCLIP, linear probe	supervised probe	0.7643	0.7473–0.7803	2-D encoder, 32 axial slices, mean-pooled ^b
GreenRFM, local adaptation	supervised head	0.7477	0.7310–0.7637	best of 10 saved checkpoints (epoch 9)
Merlin, linear probe	supervised probe	0.7144	0.6951–0.7313	abdominal/pelvic CT model ^c
MPS-CT, local reproduction	zero-shot prompt	0.6961	0.6840–0.7089	best of 12 saved checkpoints (update 3,000)
GreenRFM, local reproduction	zero-shot prompt	0.6620	not estimated	prompt readout is ill-posed for this model ^d
<i>Released adult weights, no pediatric training</i>				
BiomedCLIP	zero-shot prompt	0.6938	0.6695–0.7108	fixed prompt pair, no tuning ^b
H-LIP	zero-shot prompt	0.6916	0.6763–0.7048	released CT-RATE weights
CT-CLIP	zero-shot prompt	0.5478	0.5277–0.5670	best of the four templates in the released code ^e
CT-CLIP VocabFine	zero-shot prompt	0.5339	0.5113–0.5572	second released CT-CLIP variant ^e

^a Three-seed historical campaign (Table 10), reported with seed standard deviation rather than a bootstrap interval. All other rows are single trained models scored with identical metric code on the identical split.

^b BiomedCLIP is a 2-D biomedical vision–language model applied slice-wise with no inter-slice context. It reaches 0.503 (chance) on lung nodule against 0.935 on pleural effusion. Its probe regularization was chosen by patient-grouped cross-validation on the pediatric training split alone.

^c Merlin is an abdominal/pelvic CT foundation model. This row is out-of-domain on both body region and age, and its preprocessing resamples to 3 mm slices.

^d GreenRFM’s contrastive objective uses unnormalized dot products at fixed temperature, so per-volume embedding norm enters the cross-volume ranking. Its published result is the supervised head above.

^e The four CT-CLIP prompt templates span 0.528–0.548. A supervised probe on CT-CLIP’s own latents reaches 0.609.

Merlin’s 1,024 tokens truncate 5 of 1,507 reports. The 512 tokens of CT-CLIP and H-LIP truncate about 107. The 128 tokens that GreenRFM and MPS-CT train with truncate 1,368 of 1,507, or 91%. The median report is 267 tokens long. Extending their context would change the architecture we set out to reproduce, so we report them at their published length and treat their report→image numbers as bounded by it. Image→image retrieval never touches the text encoder and is unaffected, which is consistent with MPS-CT being a close second there while its recall is not.

9 Is the Effect Specific to Pediatrics?

The same architecture was trained and evaluated on the adult CT-RATE benchmark, using the official validation split and the full 46,394-volume training cache. It improves there too. On one final checkpoint, read from a single forward pass, the image-only general prompt branch reaches 0.8733 macro AUROC and the conditioned prompt branch reaches 0.8781 with the true indication, a paired gain of +0.0048. With every indication blanked the

same branch reaches 0.8754. The mechanism is therefore not a pediatric artifact.

The causal picture differs sharply. Blanking removes 0.27 points of a 0.48-point adult gain. In children it removes most of a gain that is an order of magnitude larger. Adult CT-RATE indications are frequently missing or generic, so most of the adult gain arises from class-specific residual adaptation rather than from indication semantics. Pediatric indications are where patient-specific clinical context actually carries diagnostic information, and that is where conditioning on it pays off. The adult numbers above are a single checkpoint. Their three-seed replication and the shuffled-indication control are registered in the companion adult paper.

10 Discussion

The pediatric improvement is large, seed-consistent, positive for 22 of 23 pathologies in the frozen snapshot and all 23 in the historical replication, positive in every pre-specified subgroup, replicated across two cohorts and two training recipes, and nearly eliminated by assigning the wrong indication. It survives the two controls most likely to expose a shortcut: the model exceeds a text-only head

Table 17: Pediatric cross-modal retrieval on the historical cohort (1,507 studies, 648 patients), no retrieval-specific fitting. Report→image is exact-match Recall@K over all 1,507 studies. Image→image is mean label-set Jaccard of the top- K retrieved studies (gallery = studies with ≥ 1 positive label): *clean* excludes the query from its own gallery; *Table-4 convention* reproduces the CT-CLIP evaluation script used by prior work, which includes the query and accumulates across K , and is shown only for comparability with published adult tables. Our arms are three-seed means; chance Recall@K is K/N .

Method	Report→image R@K				Image→image, clean			Image→image, Table-4 conv.		
	@5	@10	@50	@100	@5	@10	@50	@5	@10	@50
<i>Adapted to the pediatric cohort</i>										
METAQ-CT (full, metadata-conditioned) ^m	9.5±0.4	15.5±0.2	39.9±0.6	51.7±1.4	32.2±0.6	30.9±0.6	27.7±0.4	67.7	57.5	50.4
METAQ-CT / CT-only control (metadata-free)	10.8±0.6	17.1±0.1	41.2±0.4	54.2±1.2	31.7±0.2	30.4±0.3	27.5±0.1	67.4	57.2	50.1
GreenRFM, local adaptation (epoch 9)	0.3	1.1	5.4	9.4	27.8	26.2	23.4	66.0	54.9	47.4
MPS-CT, local reproduction (update 3,000)	6.3	10.7	29.1	42.6	30.9	28.4	24.2	67.3	56.5	48.8
<i>Released weights, no pediatric training</i>										
H-LIP	3.6	5.8	14.9	22.4	23.7	22.6	20.8	64.2	52.7	45.0
BiomedCLIP (2-D, slice-wise)	1.9	3.3	11.8	17.4	25.3	23.8	21.0	64.9	53.5	45.8
Merlin (abdominal)	1.0	1.7	6.2	10.9	21.8	21.3	19.9	63.4	51.7	44.1
CT-CLIP	0.9	1.3	5.4	10.2	19.8	19.4	18.8	62.6	50.6	43.0
CT-CLIP VocabFine	0.3	0.9	3.7	6.7	20.8	20.0	19.1	62.9	51.0	43.4
Chance (K/N)	0.3	0.7	3.3	6.6	–	–	–	–	–	–

^mUses each gallery study’s own indication, age, and sex through the conditioned bank; not an equal-input comparison with the metadata-free rows, shown to test whether conditioning changes retrieval. Every model encodes the identical impression-first report string; each uses its own tokenizer at its own trained context length, which truncates long reports: ours 256 tokens; CT-CLIP 512 (107 of 1,507 truncated); VocabFine 512 (107 of 1,507 truncated); H-LIP 512 (106 of 1,507 truncated); GreenRFM 128 (1,368 of 1,507 truncated, 91%); MPS-CT 128 (1,368 of 1,507 truncated, 91%); BiomedCLIP 256 (846 of 1,507 truncated, 56%); Merlin 1,024 (5 of 1,507 truncated). Single-model rows carry 1,000-draw patient-bootstrap intervals in the accompanying JSON.

Table 18: Readout ablation for METAQ-CT on the same pediatric retrieval task and the same stored latents as Table 17; three-seed means. The two tasks have opposite winners: the metadata-free readout retrieves the exact report best, the conditioned readout retrieves the most label-similar studies best.

Readout	R@5	R@100	Jaccard@5	Jaccard@50
z_{gen} , metadata-free (shared with CT-only)	10.8±0.6	54.2±1.2	31.7±0.2	27.5±0.1
$\text{norm}(z_{\text{gen}}+z_{\text{ind}})$, full model ^m	9.5±0.4	51.7±1.4	32.2±0.6	27.7±0.4
z_{ind} , conditioned only ^m	8.0±1.0	47.2±1.1	33.4±0.7	28.0±0.6

^mUses each gallery study’s own indication, age, and sex; not an equal-input comparison with the metadata-free row.

that already reads the indication, and its gain is larger on studies whose indication does *not* name the finding. The pattern supports a clinically plausible mechanism. Pediatric indications are rich. They carry oncologic history, surgical context, transplant status, and genetic diagnoses. They shift the prior for exactly the findings whose per-class gains are largest: metastases, post-treatment change, cysts, pneumothorax. The design lets that prior act on the pathology slots it concerns while leaving the general representation untouched.

Two further results have practical weight for anyone building pediatric CT models. The age factorial shows that a conditioned model should be trained on the full 0–18 range: only the CT-only baseline benefits from excluding the youngest children. The benchmark also shows that adult CT foundation models, as released, are not a safe starting point for pediatric deployment. Two of them are near chance, and a generic 2-D encoder with a linear layer beats the volumetric models we could reproduce.

11 Limitations

The cohort is retrospective and from a single children’s hospital. The primary frozen < 18 results are from an interim checkpoint snapshot taken while training continued. The converged three-seed comparison, the < 18 indication intervention, and the completed component matrix are queued and will replace the interim tables through the same generated fragments. The frozen test pool, while MRN-disjoint and never used for selection in this campaign, was visible during earlier exploratory analyses, so we do not claim a prospective test. The historical replication cohort is a held-out validation set, not an independent test. The conditioned branch improves discrimination but is not yet better calibrated than CT-only, so a deployed version would require post-hoc calibration. Labels are report-derived. Two of 1,507 historical validation volumes are stored non-axially or carry inconsistent header spacing and are therefore anatomically wrong input for every row of the benchmark, including ours. At 0.13% they do not move a macro AUROC but are noted for completeness. Benchmark rows are local measurements rather than published pediatric results. Several are not architecture-matched, four methods could not be reproduced at all, and a pediatric fine-tune of CT-CLIP remains outstanding. Report-generation metrics and blinded radiologist review are not part of this paper.

12 Conclusion

Conditioning a volumetric CT Q-Former on indication, age, and sex through pathology-specific queries yields a reproducible 5.8-point macro-AUROC gain on a frozen

MRN-disjoint pediatric test split and a 4.6-point gain on a second pediatric cohort, positive in every age, sex, year, and site subgroup. Counterfactual interventions and no-CT controls show that the gain comes from reading the correct patient’s clinical context and fusing it with the image, not from text alone or from added capacity. A locally measured benchmark shows that adult chest-CT foundation models transfer poorly to children and that pediatric performance must be measured directly. Together these findings argue for clinical context as an explicit, auditable input to pediatric medical vision–language models.

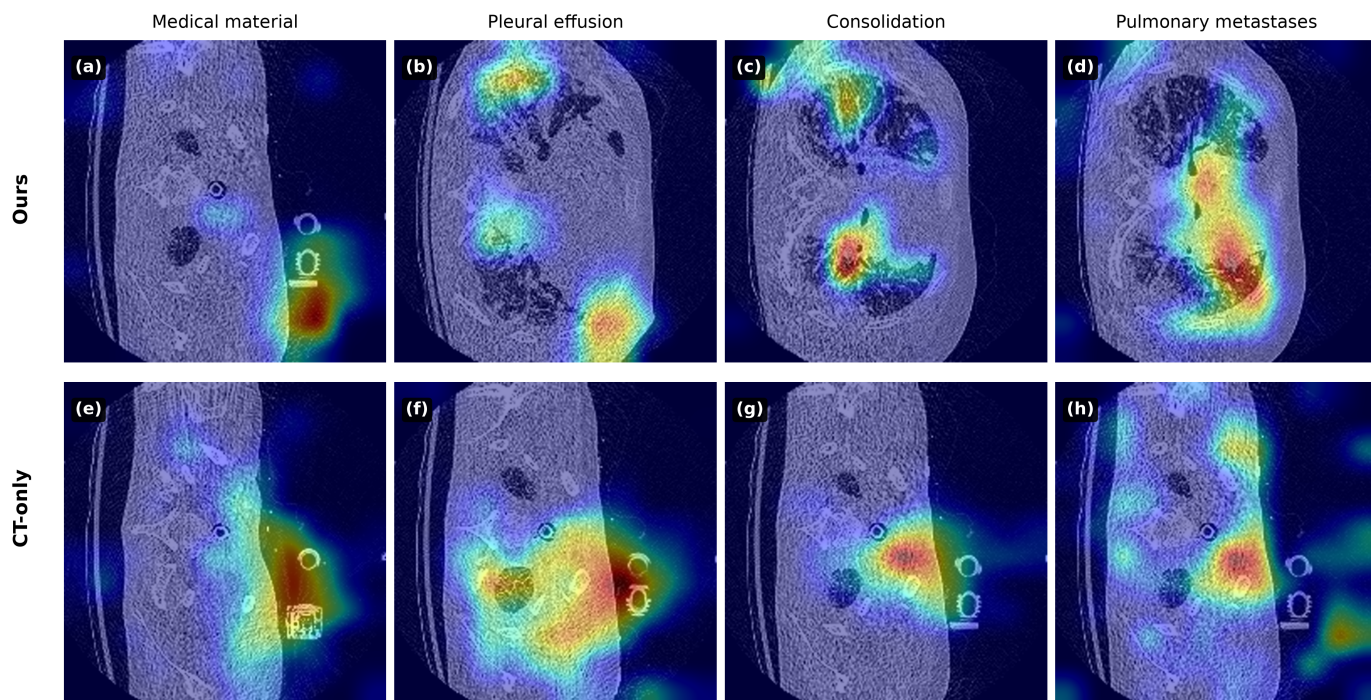


Figure 4: Qualitative class-localization comparison in the ARC-CT Figure-2 layout. Top row (a–d): METAQ-CT supplied with the correct indication, age band, and sex. Bottom row (e–h): the matched CT-only control. Columns show medical material, pleural effusion, consolidation, and pulmonary metastases. All four findings are ground-truth positive for this study. Warmer colors indicate regions with greater influence on the corresponding model score.

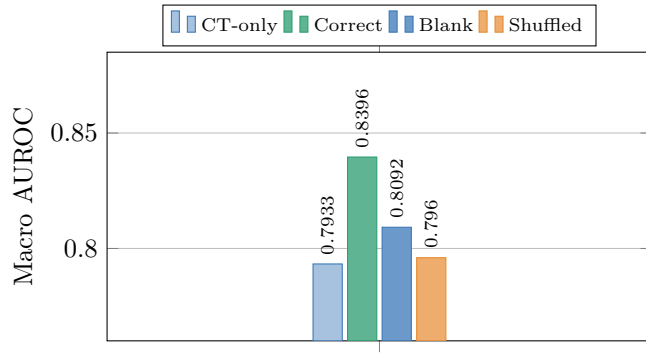


Figure 5: Indication intervention on the same trained checkpoints, historical age ≥ 5 cohort ($N = 1,507$, three-seed means). Blanking removes two-thirds of the gain over CT-only. Shuffling removes nearly all of it.

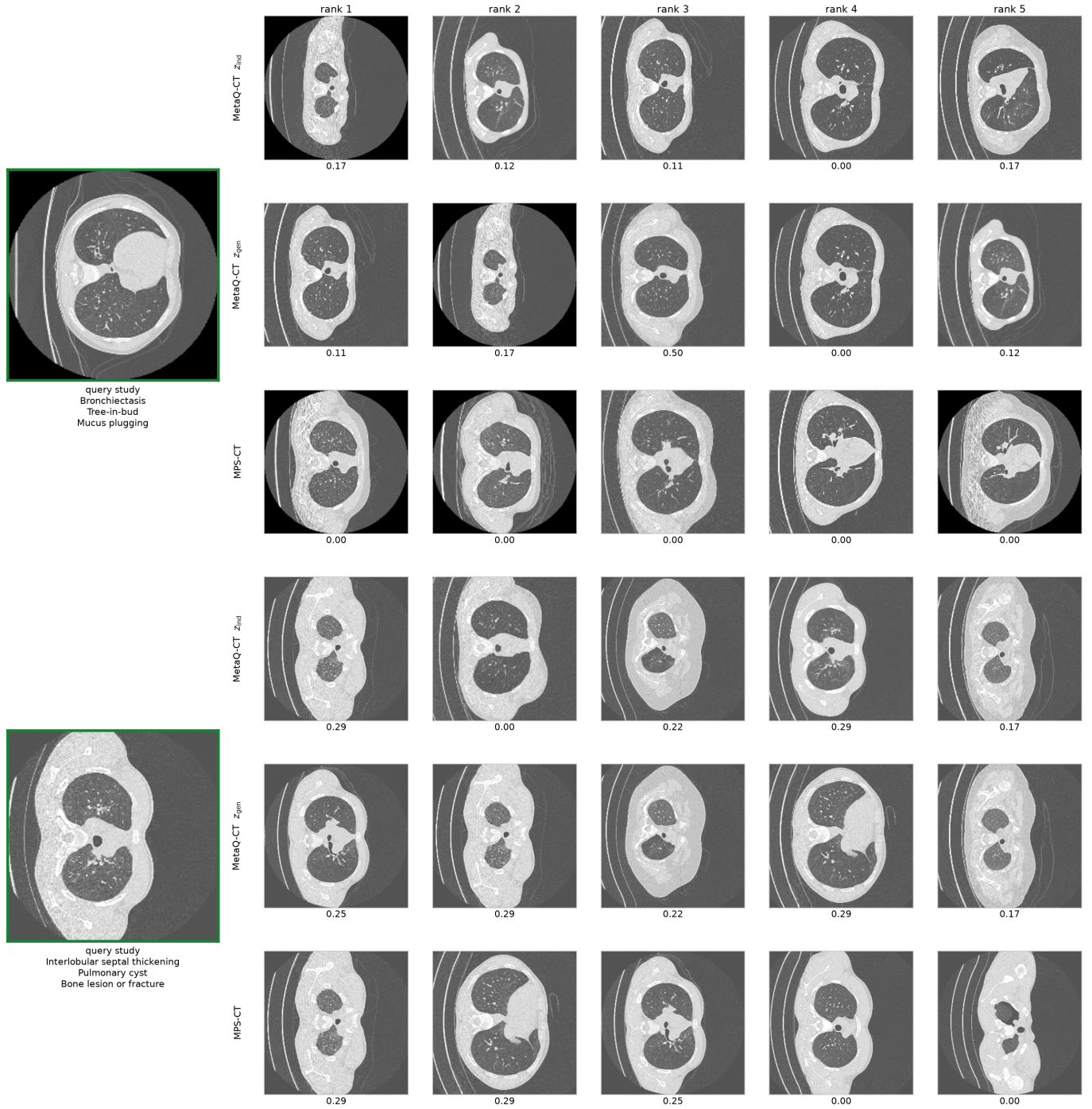


Figure 6: **Qualitative image→image retrieval.** The query study is on the left (green frame) with its ground-truth findings. Each row is one method’s top five by cosine similarity in that method’s own embedding space, with the query excluded from its own gallery. The number under each panel is the label-set Jaccard against the query, the same quantity averaged in Table 17. Queries were fixed without reference to any method’s scores: among test studies with exactly three positive findings whose field of view frames the thorax, the two whose findings are rarest in the cohort. One representative mid-thoracic axial slice is shown per study in a lung window. That slice is the one with the most lung-density voxels in the central field. Retrieval itself uses the whole volume. For the airway-disease query, MPS-CT returns studies that are globally similar in appearance but share no label with the query at any of the five ranks. This is the pediatric counterpart of the globally-similar-but-pathologically-mismatched failure reported for single-stage retrieval in prior work.

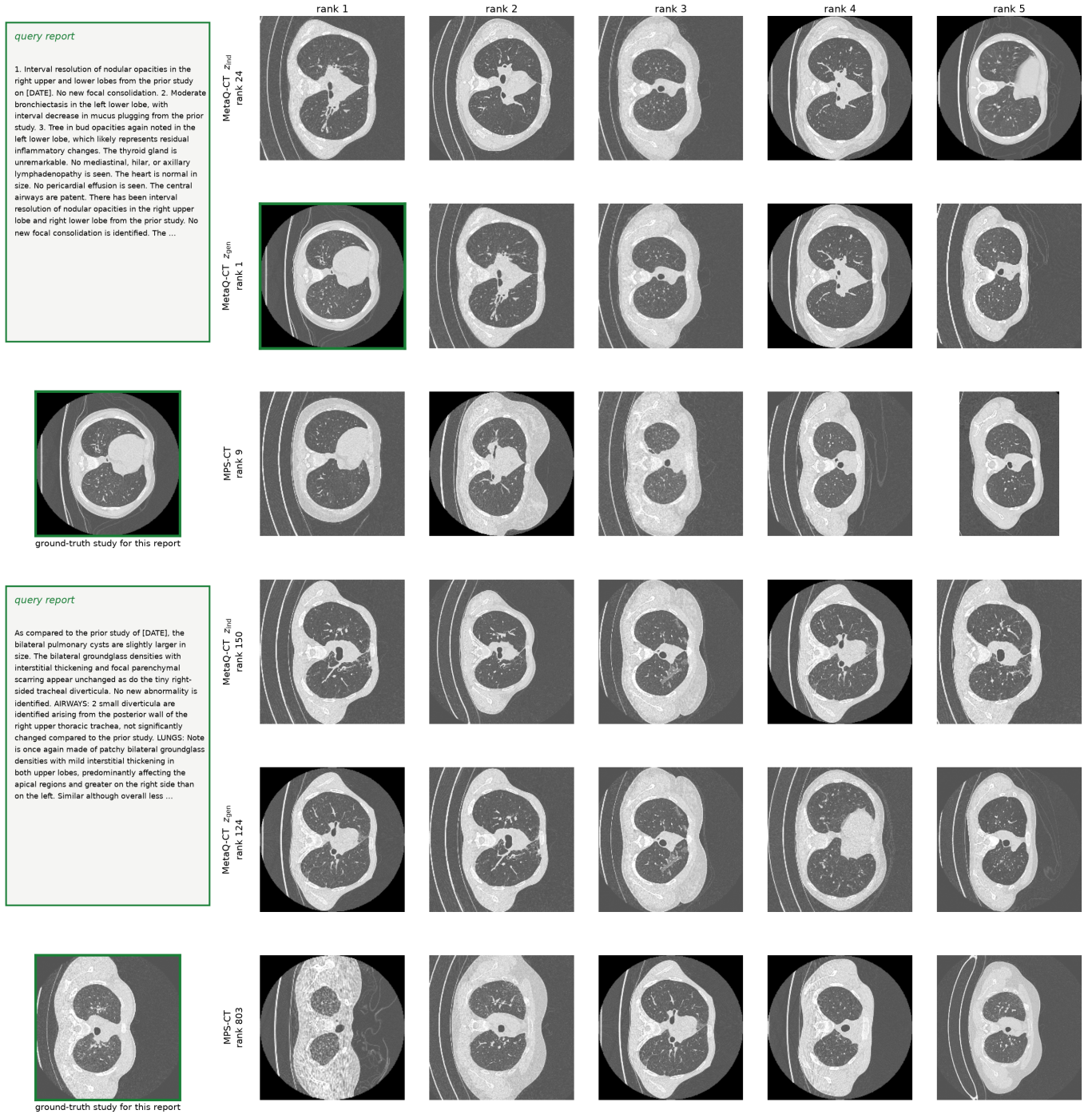


Figure 7: **Qualitative report→image retrieval.** The query is the impression-first report string, shown wrapped at the left with the study it was dictated for beneath it. The queries are the same two studies as Figure 6. Each row is one method’s top five volumes ranked by cosine similarity to that report, and the row label gives the rank the ground-truth study actually received out of 1,507. A green frame marks it when it falls inside the top five. The upper query separates the two readouts in the direction the aggregate numbers predict: the metadata-free z_{gen} places the correct study first, the conditioned z_{ind} places it 24th, and MPS-CT 9th. The lower query is a case no method solves (ranks 150, 124, and 803), included so the figure reports failures as well as successes.

A Complete Per-Class Results

The frozen < 18 per-class results are in Table 8 (generated). The matched historical age ≥ 5 per-class results are in Table 19.

B Primary-Cohort Label Distribution

Per-study prevalence of each of the 23 targets in the three splits of the primary < 18 cohort is given in Table 20. Prevalence ranges from 2.3% (pneumothorax) to 42.7% (medical material), and the development split tracks the training distribution because it was selected to do so, while the test pool was fixed independently.

C Benchmark Reproducibility Notes

Every benchmark row was re-derived from its own saved prediction file and verified against the labels file: class order identical, zero mismatched label rows, 1,507 studies from 648 patients in every case. Comparison models were run with their own released code and unmodified architectures. The only edits were to data loading and, for CT-CLIP’s trainer, making a hard-coded 18-class diagnostic list overridable so it would not crash on a 23-class cohort. Pediatric volumes are stored in Hounsfield units with the rescale carried in the NIfTI header. All pipelines were checked to apply it exactly once. The primary model starts from Kinetics initialization and pediatric Stage-1 adaptation, with no CT-RATE scan or adult ARC-CT checkpoint entering its pediatric optimization. Comparison methods instead follow the initialization prescribed by their own releases: CT-CLIP, H-LIP, Merlin, and BiomedCLIP start from released adult weights, while rows labelled “local adaptation” or “local reproduction” are additionally fitted on the pediatric training split. This initialization difference is stated explicitly because it answers a transfer question rather than a from-scratch architecture comparison.

References

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Table 19: Historical age- ≥ 5 per-class AUROC, averaged over three seeds ($N = 1,507$).

Pathology	CT-only	METAQ-CT	Δ	Pathology	CT-only	METAQ-CT	Δ
Medical material	.8562	.9024	+.0463	Peribronchial thickening	.8236	.8431	+.0195
Cardiomegaly	.9441	.9528	+.0088	Consolidation	.9241	.9286	+.0045
Pericardial effusion	.7880	.8125	+.0245	Bronchiectasis	.9101	.9250	+.0149
Lymphadenopathy	.8042	.8163	+.0121	Interlobular septal thickening	.8462	.8828	+.0366
Atelectasis	.7126	.7233	+.0107	Post-surgical/treatment change	.7485	.8658	+.1174
Lung nodule	.6507	.6937	+.0431	Pulmonary metastases	.6662	.8036	+.1374
Lung opacity	.8148	.8329	+.0181	Tree-in-bud	.8280	.8737	+.0457
Pulmonary fibrotic sequela	.7512	.7764	+.0252	Pulmonary cyst	.7523	.8548	+.1025
Pleural effusion	.9729	.9754	+.0025	Mass/neoplasm	.8496	.8680	+.0184
Mosaic attenuation	.7413	.8368	+.0956	Mucus plugging	.8622	.9014	+.0392
Pleural thickening/nodule	.6607	.6819	+.0212	Bone abnormality	.5789	.6348	+.0559
Pneumothorax	.7606	.9245	+.1639	Macro mean	.7933	.8396	+.0463

Table 20: Per-study target prevalence (%) in the primary < 18 cohort. The MRN-grouped development split was selected to preserve the multilabel distribution. The test pool was fixed independently.

Target	Train	Dev.	Test	Target	Train	Dev.	Test
Medical material	42.7	40.4	42.0	Bronchiectasis	10.8	11.9	8.1
Cardiomegaly	2.8	3.1	2.3	Interlobular septal thickening	5.6	5.3	6.0
Pericardial effusion	4.3	4.7	3.5	Post-surgical or post-treatment change	30.4	32.6	30.8
Lymphadenopathy	7.6	6.3	7.5	Pulmonary metastases	5.6	5.5	6.0
Atelectasis	34.4	34.5	35.1	Tree-in-bud	7.9	7.9	7.1
Lung nodule	48.8	48.5	49.6	Pulmonary cyst	5.3	4.7	3.9
Lung opacity	31.7	29.7	33.5	Mass or neoplasm	2.2	2.4	2.6
Pulmonary fibrotic sequela	17.9	18.4	18.5	Mucus plugging	4.5	4.7	3.5
Pleural effusion	3.8	4.0	3.8	Pleural thickening or nodule	9.8	9.5	11.5
Mosaic attenuation pattern	15.2	15.6	14.5	Bone lesion or fracture	11.1	11.1	14.1
Peribronchial thickening	15.0	15.4	13.2	Pneumothorax	2.3	1.9	2.0
Consolidation	6.5	5.1	6.0				